

Y12 Mathematics Extension 1 | Term 1 Assessment 2011

Question 1 (9 Marks)

Marks

(a) Differentiate with respect to x :

(i) $\ln(\cos^2 x)$ 2

(ii) $x^2 \operatorname{cosec}(x - 1)$ 2

(iii) $e^x \cos(3x)$ 2

(b) On a cultivated farm plot, there are approximately 12 times as many carrot plants as stinging nettle weeds. However, the plot becomes neglected and the number of weeds increases at 4% per day whilst the number of carrots decreases at 8% per day. Find the number of days, to the nearest day, that must elapse before there are more weeds than carrots. 3

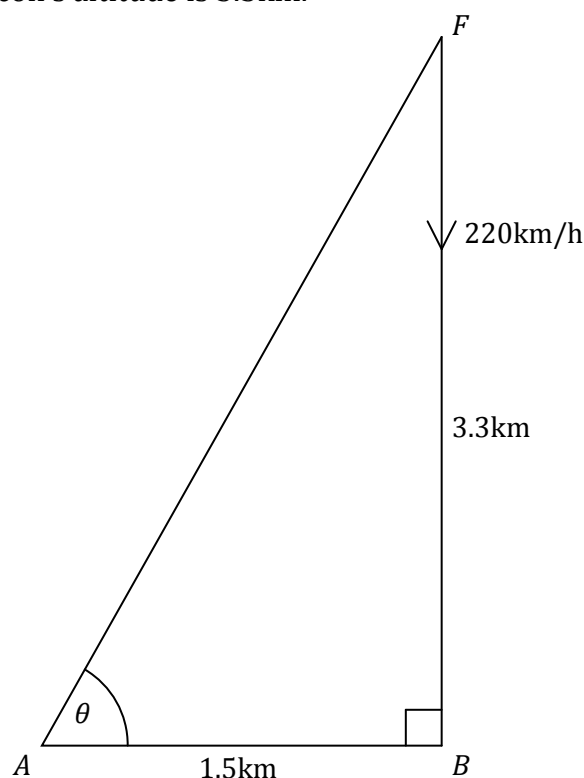
Question 2 (9 Marks) – START A NEW PAGE

Marks

(a) Find: $\int \frac{3x}{x-1} dx$. 2

(b) The area bound by $y = \sec x$, $y = 1 - 2x$, $y = 0$ and $x = \frac{\pi}{4}$ is rotated about the x -axis. Find the volume of the solid generated. 3

(c) An observer at A watches a falcon F flying downward to earth in pursuit of its prey. The falcon is descending at 220km/h and the point B directly beneath the falcon is 1.5km from the observer. Find the rate at which the observer's head is tilting, in radians per minute, when the falcon's altitude is 3.3km. 4



**Diagram
not to scale**

Question 3 (9 Marks) – START A NEW PAGE**Marks**

- (a) Express $\log \left(\frac{x^2}{\sqrt[3]{y}} \right)$ in terms of p and q if $p = \log x$ and $q = \log y$. **2**
- (b) A geometric series $S(r)$ has a first term of 5, a last term of 20,480 and a common ratio of r .
- (i) Find the difference between $S(2)$ and $S(4)$. **2**
- (ii) Find the total number of terms in $S(2)$. **1**
- (c) Neatly sketch the graph of $y = \sqrt{3} \sin x - \cos x$ for the domain $0 \leq x \leq 2\pi$, clearly showing all intercepts and stationary points. **4**

Question 4 (9 Marks) – START A NEW PAGE**Marks**

- (a) Evaluate: $\sum_{k=0}^4 3^{2k-3}$ **2**
- (b) Evaluate the following integral using the substitution $\theta = \sin^{-1} \left(\frac{x}{2} \right)$: **4**
- $$\int_0^{\sqrt{3}} \frac{2x^2}{\sqrt{4-x^2}} dx$$
- (c) Use the principle of mathematical induction to prove that $5^n + 2(11^n)$ is divisible by 3 for all positive integers n . **3**

Question 5 (9 Marks) – START A NEW PAGE**Marks**

- (a) Find: $\int \frac{x-3}{x^2+1} dx$. **2**
- (b) Find: $\int \sin 2x \sqrt{\cos 2x} dx$. **2**
- (c) The series S_n is given by the following:
- $$S_n = 3^{2k} + 3^{2k-3} + 3^{2k-6} + \dots + 3^{-k}$$
- (i) Find the value of n , the number of terms in the series, in terms of k . **1**
- (ii) Show that the n th term is given by: **1**
- $$T_n = \frac{27 \times 3^{2k}}{3^{3n}}$$
- (iii) Calculate the sum of the series. **3**

Question 6 (9 Marks) – START A NEW PAGE**Marks**

- (a) Use the principle of mathematical induction to prove that $\frac{d}{dx}x^n = nx^{n-1}$ for all positive integers n . (You may assume the product rule.) **3**
- (b) Neatly sketch the graph of $y = \ln(x^2 - 2x + 1)$. **3**
- (c) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\cot x + \operatorname{cosec}^2 x) dx$. **3**

Question 7 (9 Marks) – START A NEW PAGE**Marks**

- (a) A spherical balloon is being inflated at the rate of 85cm^3 per second. Find the rate at which its radius is increasing after one minute (3 decimal places). **3**
- (b) A man named Fischer is administered a drug that puts him to sleep. As long as there is 15mg of the drug in his body, he remains asleep. 10% of the drug leaves Fischer's body every 30 minutes. He is given an initial dose of 20mg, and an additional dose of d mg is given at the beginning of each new hour.
- (i) Show that immediately after the second additional dose is administered, Fischer has $13.122 + 1.81d$ mg of the drug remaining in his body. **1**
- (ii) If R_n is the amount of drug remaining at the end of the n th hour, just before he receives his additional dose, show that R_n is given by: **3**
- $$R_n = 20(0.81)^n + d \left[\frac{81}{19} - \frac{(0.81)^n}{0.19} \right]$$
- (iii) If Fischer must be kept asleep for at least ten hours, find the minimum hourly dose required (to the nearest 0.01mg). **2**

Suggested Solutions	Marks	Marker's Comments
$1(a)(i) \frac{d}{dx} \ln(\cos^2 x) = \frac{-2\sin x \cos x}{\cos^2 x}$ $= \frac{-2\sin x}{\cos x}$ $= -2\tan x$	1 $\frac{1}{2}$ $\frac{1}{2}$	
$1(a) \text{ OR } \frac{d}{dx} \ln(\cos^2 x) = \frac{d}{dx} (2\ln \cos x) \text{ for } 0 < x < \frac{\pi}{2}$ $= \frac{-2\sin x}{\cos x}$ $= -2\tan x$	1 $\frac{1}{2}$ $\frac{1}{2}$	
$(ii) \frac{d}{dx} (x^2 \operatorname{cosec}(x-1)) = \frac{d}{dx} \left[\frac{x^2}{\sin(x-1)} \right]^{\frac{1}{2}}$ $= \frac{\sin(x-1) \times 2x - x^2 \times \cos(x-1)}{\sin^2(x-1)}$ $\text{OR } = 2x \operatorname{cosec}(x-1) - x^2 \cot(x-1) \operatorname{cosec}(x-1)$ $\text{OR } = x \operatorname{cosec}(x-1) (2 - x \cot(x-1))$	$\frac{1}{2}$ 1 $\frac{1}{2}$	
$(iii) \frac{d}{dx} [e^x \cos(3x)] = \cos(3x)e^x + e^x(-3\sin(3x))$ $= e^x [\cos(3x) - 3\sin(3x)]$	2	1 mark each for the parts of the product rule
(b) Let t be the number of days elapsed $\therefore$ Number of weeds = $n(1.04)^t$ for $n \in \mathbb{R}$ Number of carrots = $12n(0.92)^t$ Solving $12n(0.92)^t = n(1.04)^t$ $12 = \frac{(1.04)^t}{(0.92)^t}$ $12 = \left(\frac{1.04}{0.92} \right)^t$ $12 = \left(\frac{26}{23} \right)^t$ $t = \log \left(\frac{26}{23} \right)^{12} \text{ i.e. } \frac{\ln 12}{\ln 26 - \ln 23}$ $t = 20.268022...$ $\therefore$ 21 days must elapse before there are more weeds than carrots.	1 1 1	1 mark lost for $t-1$ as the power

MATHEMATICS Extension 1 : Question 2

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{(a)} \quad \int \frac{3x}{x-1} dx & \quad \text{or} \quad \int \frac{3x-3+3}{x-1} dx \\ &= \int \left(3 + \frac{3}{x-1} \right) dx = \int \left(3 + \frac{3}{x-1} \right) dx \\ &= 3x + 3\ln(x-1) + C \end{aligned}$$

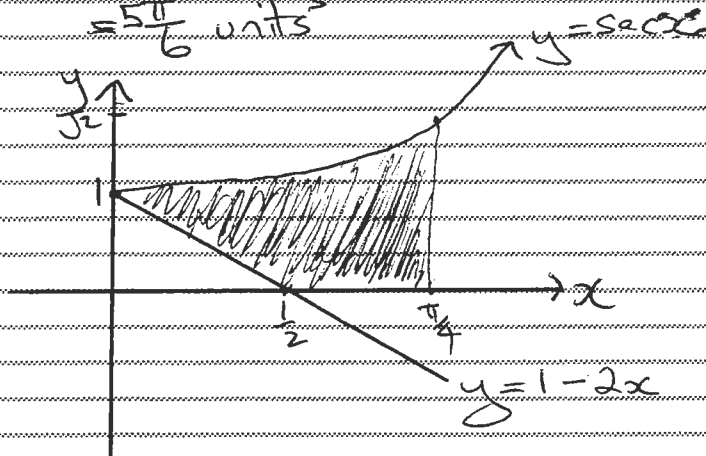
1mk

1mk

* generally well done.
(* Some did it by substitution).

$$\begin{aligned} \text{(b)} \quad V &= \pi \int_0^{\pi/4} \sec^2 x dx - \frac{1}{3} \pi \times (1)^2 \times \frac{1}{2} \\ &= \pi [\tan x]_0^{\pi/4} - \frac{\pi}{6} \\ &= \pi (\tan \frac{\pi}{4} - \tan 0) - \frac{\pi}{6} \\ &= \pi (1 - 0) - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \text{ units}^3 \end{aligned}$$

$$\begin{aligned} \text{or} \quad V &= \pi \int_0^{\pi/4} \sec^2 x dx - \pi \int_0^{1/2} (1-2x)^2 dx \\ &= \pi [\tan x]_0^{\pi/4} - \pi \int_0^{1/2} (1 + 4x^2 - 4x) dx \\ &= \pi (\tan \frac{\pi}{4} - \tan 0) - \pi \left[x - 2x^2 + \frac{4}{3}x^3 \right]_0^{1/2} \\ &= \pi (1 - 0) - \pi \left(\frac{1}{2} - (2 \times \frac{1}{4}) + (\frac{4}{3} \times \frac{1}{8}) \right) \\ &= \pi - \pi \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) \\ &= \frac{5\pi}{6} \text{ units}^3 \end{aligned}$$



* lost 1/2 mk if they mixed up the cylinder's height and radius.
* 1/2 mk off if they couldn't subtract properly.

* If they did $\int_0^{\pi/4} (1-2x)^2 dx$ then they lost 1/2 mk.

* no marks for graphing it correctly.

* If they only did $V = \pi \int_0^{\pi/4} \sec^2 x dx = \pi$ they got 1mk.

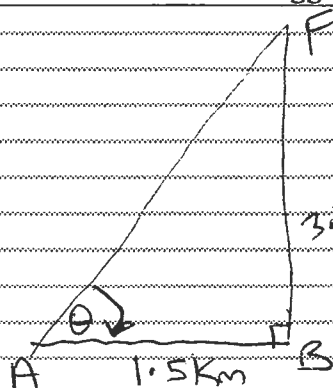
* If they did the area only and made errors they got 0

MATHEMATICS Extension 1 : Question 2 cont.

Suggested Solutions

Marks

Marker's Comments



let falcon's altitude = x

$$\tan \theta = \frac{x}{1.5}$$

$$x = 1.5 \tan \theta$$

$$\frac{dx}{d\theta} = 1.5 \sec^2 \theta = \frac{3}{2 \cos^2 \theta}$$

now $\frac{dx}{dt} = -220$ (falcon is flying down)

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$$

$$= \frac{2 \cos^2 \theta}{3} \times -220$$

when $x = 3.3$, $\tan \theta = \frac{3.3}{1.5}$

$$\theta = 1.144168834$$

$$\therefore \frac{d\theta}{dt} = \frac{-5500}{219}$$

$$\therefore \frac{d\theta}{dt} = \frac{-5500}{219 \times 60} = \frac{-275}{657}$$

$\therefore$ The observer's head is tilting downwards at $\frac{275}{657}$ rad/min.

or $\tan \theta = \frac{x}{1.5}$

$$\theta = \tan^{-1} \left(\frac{x}{1.5} \right)$$

$$\frac{d\theta}{dx} = \frac{1.5}{1.5^2 + x^2}$$

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{d\theta}{dx} \times \frac{dx}{dt} \\ &= \frac{1.5}{1.5^2 + x^2} \times -220 \\ &= \frac{-330}{1.5^2 + x^2} \end{aligned}$$

$\frac{1}{2}$ mk

* If they get $\frac{dx}{dt} = 220$ only, they lost 1mk; interpretation made question easier.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

* If they left it in degrees, lost marks.

* If they left it in rad/hr, lost $\frac{1}{2}$ mk.

when $x = 3.3$,

$$\frac{d\theta}{dt} = \frac{-5500}{219} \text{ rad/hr}$$

$$\frac{d\theta}{dt} = \frac{-275}{657} \text{ rad/min}$$

$\therefore$ The observer's head

----- etc.

Y12 MATH EXT ASSESSMENT TASK 2
TERM 1, 2011

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

Q 3 (a)

$$\log \frac{x^2}{\sqrt[3]{y}} = \log x^2 - \log \sqrt[3]{y} ; x, y > 0$$

$$= 2 \log x - \log y^{\frac{1}{3}} = 2 \log x - \frac{1}{3} \log y$$

$$= 2p - \frac{1}{3}q$$

$\frac{1}{2}$ for -
 $\frac{1}{2}$ for 2
 $\frac{1}{2}$ for $\frac{1}{3}$
 $\frac{1}{2}$ for answer

[2]

(b) $T_1 = a = 5$

(i) $S(r) = 5 + 5r + 5r^2 + 5r^3 + \dots + 5r^{n-1} = \frac{5(r^n - 1)}{r - 1}$

$= \frac{r^n - a}{r - 1} = \frac{r^n - 5}{r - 1}$ $\frac{1}{2}$

For $S(2) = 5 + 5 \cdot 2 + 5 \cdot 2^2 + 5 \cdot 2^3 + \dots + 20480$ 5×2^{12}
 $S(4) = 5 + 5 \cdot 4 + 5 \cdot 4^2 + 5 \cdot 4^3 + \dots + 20480$ 5×4^6

So $S(2) = \frac{5(2^n - 1)}{2 - 1} = 2 \times 4 - 5 = 2 \times 20480 - 5 = 40955$ $\frac{1}{2}$

$S(4) = \frac{5(4^n - 1)}{4 - 1} = \frac{4 \times 4 - 5}{3} = \frac{4 \times 20480 - 5}{3} = 27305$ $\frac{1}{2}$

$\therefore S(2) - S(4) = 40955 - 27305 = 13650$ $\frac{1}{2}$

Acc - 13650

OR $S(2) - S(4) = 5 \cdot 2 + 5 \cdot 2^3 + 5 \cdot 2^5 + 5 \cdot 2^7 + 5 \cdot 2^9 + 5 \cdot 2^{11}$
 $= 5 \cdot 2 [1 + 2^2 + 2^4 + 2^6 + 2^8 + 2^{10}]$
 $= 10 \times [(2^5)^2 - 1] = 13650$

[2]

(ii) As a Geom. series $T_n = ar^{n-1}$ $n=1$

So $20480 = 5 \times 2$

$2^{n-1} = 4096 = 2^{12}$

$\therefore n = 13$

$\therefore$ No of terms in $S(2)$ is 13

[1]

(c) $y = \sqrt{3} \sin x - \cos x$
 $= 2 \sin(x - \frac{\pi}{6}) ; 2 \sin(x + \frac{11\pi}{6}) ; 2 \cos(x - \frac{2\pi}{3}) ; 2 \cos(x + \frac{4\pi}{3})$

$R=2$ and
Correct phase $\angle (1)$

$(0, -1) \dots (\frac{1}{2}) (2\pi, -1)$

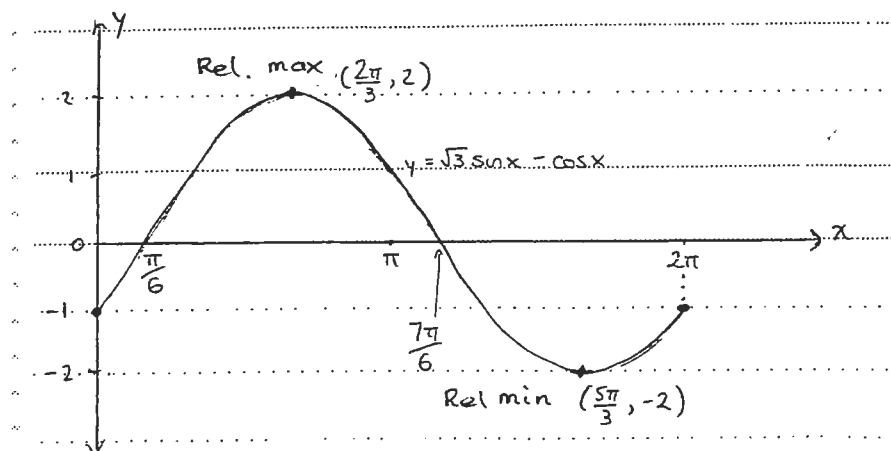
x -ints $\frac{\pi}{6}$ and $\frac{7\pi}{6}$ (1)

SPs $\left\{ \begin{array}{l} (\frac{2\pi}{3}, 2) \text{ Rel max TP} \\ (\frac{5\pi}{3}, -2) \text{ Rel min TP} \end{array} \right.$

$\frac{1}{2}$ each

$\frac{1}{2}$ shape

[4]



Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 \text{a) } \sum_{k=0}^4 3^{2k-3} &= 3^{-3} + 3^{-1} + 3^1 + 3^3 + 3^5 \\
 &= \frac{1}{27} + \frac{1}{3} + 3 + 27 + 243 \\
 &= \frac{7381}{27} = 273\frac{10}{27}
 \end{aligned}$$

1

1

Either answer got full marks.

$$\begin{aligned}
 \text{b) } \theta &= \sin^{-1}\left(\frac{x}{2}\right) \Rightarrow \frac{d\theta}{dx} = \frac{1}{\sqrt{4-x^2}} \\
 \therefore \frac{dx}{\sqrt{4-x^2}} &= d\theta
 \end{aligned}$$

$$\text{Also } x = 2 \sin \theta$$

$$\text{When } x = \sqrt{3}, \theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

$$x = 0 \quad \theta = \sin^{-1}(0) = 0$$

$$\text{Integral is } \int_0^{\pi/3} 2(2 \sin \theta)^2 d\theta$$

$$= 8 \int_0^{\pi/3} \sin^2 \theta d\theta$$

$$= \frac{8}{2} \int_0^{\pi/3} (1 - \cos 2\theta) d\theta$$

$$= 4 \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/3}$$

$$= 4 \left(\frac{\pi}{3} - \frac{\sqrt{3}/2}{2} \right) - 4(0 - 0)$$

$$= \frac{4\pi}{3} - \sqrt{3}$$

 $\frac{1}{2}$

(limits)

1

1

1

 $\frac{1}{2}$

Most people transferred to $x = \sin 2\theta$ before finding $d\theta$

This produced

$\sqrt{\cos^2 \theta}$ in the denominator.

Only a few explained away $|\cos \theta|$ correctly (or even noticed it !)

Suggested Solutions

Marks

Marker's Comments

c) To Prove $5^n + 2(11^n)$ is divisible by 3 for integer $n \geq 1$.

Step 1 If $n=1$, $5^n + 2(11^n) = 5 + 22$
 $= 27$
 $= 3 \times 9$

Thus true for $n=1$.

Step 2 Assume result is true for some integer $k \geq 1$.

i.e. $5^k + 2(11^k) = 3A$, some integer A . *

It is necessary to prove that

$5^{k+1} + 2(11^{k+1})$ is divisible by 3.

$$\begin{aligned} 5^{k+1} + 2(11^{k+1}) &= 5 \times 5^k + 2 \times 11 \times 11^k \\ &= 5 \times 5^k + 22 \times 11^k \\ &= 5(5^k + 2(11^k)) + 12 \times 11^k \\ &= 5 \times 3A + 12 \times 11^k \quad (\text{By assumption}^*) \\ &= 3(15A + 4(11^k)) \end{aligned}$$

which is divisible by 3. Thus, if true for k , then also true for $k+1$.

Step 3 Using steps 1 and 2, we can deduce by P.M.I. that

$5^n + 2(11^n)$ is divisible by 3 for integer $n \geq 1$.

$\frac{1}{2}$

2

$\frac{1}{2}$

Essentially 2 marks for a correctly argued step 2 plus 1 mark for the structure, including a correct step 1 and step 3.

Step 3 was generally weak.

Although most people knew the general idea, it was quite poorly presented.

MATHEMATICS Extension 1 : Question.....5

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{(a)} \int \frac{x-3}{x^2+1} dx &= \int \frac{x}{x^2+1} dx - 3 \int \frac{dx}{x^2+1} \\ &= \frac{1}{2} \ln(x^2+1) - 3 \tan^{-1} x + C \end{aligned}$$

4

1 each

$$\text{b)} \int \sin 2u (\cos 2u)^{1/2} du = -\frac{1}{3} (\cos 2u)^{3/2} + C$$

2

1 est each error

$$\begin{aligned} \text{(c)} \text{(i)} S_n &= 3^{2k} + 3^{2k-3} + 3^{2k-6} + \dots + 3^{-k} \\ T_n &= 3^{2k} \cdot \left(\frac{1}{3}\right)^{n-1} \\ &= 3^{2k-3n+3} \end{aligned}$$

1

$$\begin{aligned} \text{Number terms } 2k-3n+3 &= -k \\ 3n &= 3k+3 \\ n &= k+1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} T_n &= 3^{2k} \left(\frac{1}{3}\right)^{n-1} \\ &= 3^{2k} \cdot \frac{1}{3^{3n-3}} \\ &= 27 \cdot 3^{2k-3n} \end{aligned}$$

1

$\frac{1}{2}$ est each mistake.

$$\begin{aligned} \text{(iii)} S_n &= 3^{2k} + 3^{2k-3} + 3^{2k-6} + \dots + 3^{-k} \\ 3^{-3} S_n &= 3^{2k-3} + 3^{2k-6} + \dots + 3^{-k-3} \end{aligned}$$

$$(1-3^{-3})S_n = 3^{2k} - 3^{-k-3}$$

3

1 est each mistake.

$$\begin{aligned} S_n &= \frac{27}{26} \left[3^{2k} - 3^{-k-3} \right] \quad \text{OR} \\ &= \frac{3^{2k+3} - 3^{-k}}{26} \end{aligned}$$

index rules.
simplification needed.

$$\begin{aligned} \text{OR } S_n &= \frac{a(1-r^n)}{1-r} \quad n=k+1 \\ &= \frac{3^{2k} (1-3^{-3(k+1)})}{1-3^{-3}} \\ &= \frac{3^{2k+3} - 3^{-k}}{27-1} \\ &= \frac{3^{2k+3} - 3^{-k}}{26} \end{aligned}$$

MATHEMATICS Extension : Question 6

Suggested Solutions	Marks	Marker's Comments
Q6(a) Let $P(n)$ be the proposition: $\frac{d}{dx} x^n = nx^{n-1}$ for $n=1, 2, 3, \dots$ • For $P(1)$: $\begin{aligned} \text{LHS} &= \frac{d}{dx} [x] ; f(x) = x & \text{RHS} &= 1 \cdot x^{1-1} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} & &= 1 \cdot x^0 \\ &= \lim_{h \rightarrow 0} \frac{h}{h} & &= 1 \\ &= \lim_{h \rightarrow 0} 1, h \neq 0 \text{ as } h \rightarrow 0. \\ &= 1 \end{aligned}$ $\therefore$ LHS = RHS $\therefore P(1)$ is true • Assume $P(n)$ is true upto some integer $k \geq 1$ i.e. $\frac{d}{dx} x^k = kx^{k-1}$ (*) At $P(k+1)$ is true for $n = k+1$ i.e. $\frac{d}{dx} x^{k+1} = (k+1)x^k$ • <u>PROOF</u> for $P(k+1)$ $\begin{aligned} \text{Now } \frac{d}{dx} x^{k+1} &= \frac{d}{dx} [x^k \cdot x] \\ &= \frac{d}{dx} x^k \cdot x + x^k \cdot \frac{d}{dx} x \\ &= kx^{k-1} \cdot x + x^k \cdot 1 \\ &= kx^k + x^k \\ &= (k+1)x^k \end{aligned}$ $\therefore P(k+1)$ is true • Hence by the PMI $P(n)$ is true for $n=1, 2, 3, \dots$	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	(1) For $\frac{d}{dx} x = 1$ if explain via First Principles • Geometrical - $\frac{1}{2}$ if state $k \in \mathbb{Z}^+$ "$k \geq 1$" $P(k)$ Product Rule using $P(1)$ and Assumption 3
(b) $y = \ln(x^2 - 2x + 1) = \ln(x-1)^2$ $\forall x > 1$	Max 2	$2\ln(x-1)$ only if $x > 1$. 1 For $\forall x > 0$ with sketch concave downwards $\frac{1}{2}$ For each intercept $\frac{1}{2}$ For c.d slope 2 for $y = 2\ln(x-1)$ sketch. 3
(c) $I = \int_{\pi/3}^{\pi/2} (\cot x + \csc^2 x) dx = \int_{\pi/3}^{\pi/2} \left(\frac{\cos x}{\sin x} + \csc^2 x \right) dx$ $\begin{aligned} &= \left[\ln(\sin x) - \cot x \right]_{\pi/3}^{\pi/2} \\ &= \left[\ln(\sin \frac{\pi}{2}) - \cot \frac{\pi}{2} \right] - \left[\ln(\sin \frac{\pi}{3}) - \cot \frac{\pi}{3} \right] \\ &= \left[\ln 1 - 0 \right] - \left[\ln \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{3}} \right] \\ I &= \frac{1}{\sqrt{3}} - \ln\left(\frac{\sqrt{3}}{2}\right) \end{aligned}$	$\frac{1}{2}$	1 for each $\frac{1}{2}$ For substit. answer (correct) 3

Suggested Solutions	Marks	Marker's Comments
a) $\frac{dr}{dt} = \frac{dr}{dv} \times \frac{dv}{dt} = \frac{1}{4\pi r^2} \times 85$ $t=60 \quad V = 60 \times 85 = 5100 = \frac{4}{3}\pi r^3$ $r = \left(\frac{3825}{\pi}\right)^{\frac{1}{3}} \doteq 10.67809 \dots$ $\frac{dr}{dt} = \frac{85}{4\pi(10.67809)^2} = 0.5932 \dots$ radius increasing at 0.59 cm/sec or $\underline{3.559 \text{ cm/min}} \#$	1m 1m 1m	many get r wrong
b) After 1st additional dose = $20(0.9)^2 + d$ After 2nd additional dose = $[20(0.9)^2 + d] \cdot 0.9 + d$ $\underline{= 13.122 + 1.81d} \#$	1m	
ii) Before additional dose $R_1 = 20(0.81)$ $R_2 = 20(0.81)^2 + d(0.81)$ $R_3 = 20(0.81)^3 + d(0.81^2 + 0.81)$ $\vdots$ $R_n = 20(0.81)^n + d(0.81^{n-1} + 0.81^{n-2} + \dots + 0.81)$ $= 20(0.81)^n + d \frac{0.81}{0.19} (1 - 0.81^{n-1})$ $= 20(0.81)^n + d \left[\frac{81}{19} - \frac{0.81^n}{0.19} \right] \#$	1m 1m 1m 1m	many fudging if show consistency in GP get 1m must show the first term & last term, min 3 terms
iii) $R_{10} = 15 \quad 20(0.81)^{10} + d \left(\frac{81}{19} - \frac{0.81^{10}}{0.19} \right) = 15$ $d = \frac{15 - 20(0.81^{10})}{\frac{81}{19} - \frac{0.81^{10}}{0.19}} = 3.47 \text{ mg} \#$ (repeat 0.01 mg)	1m 1m	Students solve $R_{10} = 0$ Get no marks as it makes calculation easier & the answer is -0.67 mg does not make sense